

DISHA POTRU

DATA ANALYST

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SUMMARY

Results-driven Data Analyst with 3 years of experience in data analysis and a strong specialization in extracting, integrating, and analyzing complex datasets from pharmacy management systems and financial claims databases. Proficient in Python, SQL, and Java, with expertise in advanced Excel functions and data visualization tools such as Tableau and Power BI to facilitate data-driven decision-making. Experienced in utilizing Amazon Web Services (AWS) for effective data storage and processing, with strong skills in SQL databases, including MySQL, PostgreSQL, MongoDB, and SQL Server. Demonstrated ability to implement predictive models and data mining techniques, leveraging machine learning libraries to optimize pharmacy inventory levels and enhance revenue through targeted marketing initiatives.

SKILLS

Methodologies:	SDLC, Agile (Scrum, Kanban), Waterfall
Languages:	Python, SQL, R, SAS, Java
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ
Packages:	NumPy, Pandas, Matplotlib, SciPy, ggplot2, TensorFlow, Seaborn, Scikit-learn
Visualization Tools:	Tableau, Power BI, Advanced Excel, Statistics
Database:	MySQL, MongoDB, PostgreSQL, SQL Server, Oracle
Other Skills:	Amazon Web Services (AWS), ETL, SSIS, SSRS, EDA, Machine Learning Algorithms, Probability distributions, Confidence Intervals, ANOVA, Hypothesis Testing, Regression Analysis, Linear Algebra, Advanced Analytics, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Storytelling, Association rules, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, Jira, Git, GitHub, SVN
Operating System:	Windows, Linux, Mac OS
Soft Skills:	Time Management, Leadership, Strategy Planning, Problem-Solving, Negotiation, Decision-Making, Documentation and Presentation, Analytical Thinking, Attention to Detail, verbal, and written communication

EDUCATION

Master of Science in Computer and Information Sciences | Tampa, Florida, USA

Saint Leo University

Bachelor of Technology | Hyderabad, India,

G.Narayanamma Institute of Technology and Science,

EXPERIENCE

Data Analyst | [HCA Healthcare](#), FL | Aug 2024 – Present

- Initiated and managed a project under Agile methodology, collaborating with clinical teams and data governance stakeholders to define project scope, compliance requirements (HIPAA, PHI), and key performance indicators (KPIs) for chronic disease prediction and readmission risk.
- Built scalable ETL pipelines utilizing SSIS, SQL Server, and Python to extract and clean both structured (claims, vitals, medication adherence) and semi-structured data (lab reports, encounter summaries), achieving data quality of 97% through validation checks.
- Conducted in-depth data profiling and analysis using Pandas, NumPy, and SQL, identifying key predictors of readmission and segmenting high-risk populations for enhanced clinical insights.
- Developed and validated predictive models using scikit-learn, applying decision trees and logistic regression, achieving an AUC of 0.86 with 82% accuracy in flagging high-risk patients.
- Implemented NLP techniques with SpaCy and NLTK to extract behavioral patterns, comorbidities, and patient sentiment from unstructured physician notes, enhancing model recall by 18%.
- Created interactive Tableau dashboards to monitor KPIs such as readmission risk scores and ER visit trends, allowing for drill-down filtering to facilitate data-driven departmental decision-making.

- Deployed predictive models and dashboards on Azure, integrating with internal workflows using APIs and JIRA task boards to enhance alerting for care teams.
- Achieved a 40% reduction in reporting time and a 30% increase in patient outreach speed through automation, resulting in significant cost savings and improved patient engagement.

Data Analyst | [Deloitte](#), India | May 2021 – July 2023

- Initiated and managed a financial forecasting project using Waterfall methodology, collaborating with finance stakeholders to gather business requirements for quarterly cost forecasting, burn rate tracking, and variance analysis.
- Engineered scalable data pipelines with Apache Spark to extract and process high-volume financial data, including expense transactions and journal entries, enhancing cost and budget analytics.
- Employed Python for data normalization, missing value imputation, and spend type classification, ensuring data consistency by ingesting from MongoDB and AWS S3 for downstream financial modeling.
- Conducted statistical modeling and variance analysis using ANOVA and multiple linear regression in Jupyter Notebook, which led to early detection of 20%+ overspending in two business units.
- Developed and deployed time-series cost forecasting models in Python to predict quarterly OPEX and CAPEX performance, achieving >85% forecast accuracy and providing actionable insights to FP&A teams.
- Designed interactive dashboards in Power BI to visualize key financial metrics such as department-level spend, monthly variances, and actuals vs. forecast, enhancing accessibility for finance leads and executives.
- Utilized JIRA for effective issue tracking and milestone monitoring, further supporting dashboarding efforts with Advanced Excel reports including PivotTables and Power Query for executive reviews.
- Delivered measurable impacts by helping the finance division reallocate underutilized budgets, reduce excess spending by ~15% across projects, and streamline quarterly planning through automated data pipelines.

PROJECTS

- **Project: Generative AI Chatbot with LangChain and VectorDB** | [Python](#), [LangChain](#), [OpenAI GPT](#), [ChromaDB](#), [Streamlit](#). Built an end-to-end Generative AI application capable of answering user queries based on uploaded document context using Retrieval-Augmented Generation (**RAG**). Utilized **LangChain** to orchestrate the pipeline—loading documents, chunking, embedding, and retrieval. Employed **OpenAI's** GPT model for contextual responses and **ChromaDB** for efficient vector similarity search. Integrated a **Streamlit**-based frontend to enable real-time interaction, query input, and response visualization. Enabled scalable, context-aware document-based question answering, demonstrating the potential of LLMs in enterprise knowledge automation.
- **Project: AI-Powered Sentiment Analysis for Social Media** | [Python](#), [Machine Learning](#), [AWS](#), [NLP](#). Designed and deployed a cloud-based sentiment analysis solution on AWS to classify social media text into positive, negative, neutral, and mixed categories. Implemented advanced NLP techniques and machine learning models to process unstructured data in real time, enabling businesses to extract actionable insights that enhance marketing strategies, customer engagement, and brand reputation management.